

Sparsh Chakraborty

New Delhi, India
LinkedIn

Email: sparsh.chakraborty07@gmail.com
Mobile: +91-7290947005
GitHub: @codexx07

EDUCATION

- Vellore Institute of Technology, Chennai Campus** Chennai, India
 - Bachelors of Computer Science Engineering, Specialization in AI & ML; CGPA: 9.08/10* Sep 2022 - May 2026
 - Courses: Data Structures, Algorithm Design, Machine Learning, Deep Learning, Machine Vision, Cryptography and Network Security, Operating Systems, Computer Networks, Database Systems, Speech and Language Processing, Autonomous Drones*

SKILLS SUMMARY

- Languages:** Python, C++, Java, JavaScript, SQL
- Frameworks:** Pytorch, Tensorflow, CUDA, Docker, ReactJS, Django, Vue.js, FastAPI
- Tools:** Git, MATLAB, Keil uVision, LTspice
- Spoken Languages:** English (Full Proficiency), Hindi (Native), Bengali (Native)

EXPERIENCE

- Summer Research Intern – ASR & Speech Enhancement** Samsung Research
 - May 2025 – Jul 2025* Bangalore, India
 - End-to-End ASR Pipeline:** Designed and implemented an automatic speech recognition (ASR) system using transformer-based acoustic modeling and language models.
 - Speech Enhancement:** Built and optimized deep learning models for noise suppression and speech clarity improvement in real-world environments.
 - Previous Collaboration:** Previously working remotely under the Samsung PRISM Research Mentorship Program from Aug 2024 – Dec 2024, focusing on audio restoration and real-time model optimization.
- AI and Robotics Intern** StemTec Pvt Ltd (VIT Chennai Incubated)
 - Feb 2024 - Aug 2024* Chennai, India
 - Leadership:** Directed a team of 5, integrating computer vision models into AGV prototypes using RealSense SDK and PyTorch Transformers.
 - ROS2 Integration and Semantic Segmentation:** Integrated computer vision models into AGV prototypes via ROS2, ensuring optimal performance.
- Software Development Intern** Stair Digital Pvt Ltd
 - Sep 2023 - Feb 2024* Remote
 - Web Development:** Developed responsive web applications using React, Django, and PostgreSQL, enhancing user engagement.
 - Data Extraction Automation:** Automated web scraping processes, enabling efficient data analysis and supporting strategic decision-making.

PROJECTS

- MRIG (Healthcare System Integration):** Collaborated with cross-functional teams to integrate AI models into healthcare systems, improving system interoperability by 35%. **GitHub.** Tech: Python, TensorFlow, Docker
- VitaFile (Google APIs Integration):** Spearheaded the development using Google APIs to centralize EHR access. Innovated user experience with multilingual support, voice interaction, and gesture recognition. **GitHub.** Tech: Python, Google APIs, React, Django
- Ollash (t2\$sh CLI):** Designed a CLI assistant that converts natural language to safe, POSIX-compliant shell commands using local LLMs via Ollama. Supports interactive REPL, history-aware retrieval with Interro, and GGUF/Hugging Face model integration. **Website** Tech: Python, Ollama, Interro, Docker, CLI

RESEARCH INTERNSHIPS

- IIIM Kozhikode – Social Media & Brand Analytics:** Applied **NLP, sentiment/emotion detection, and topic modeling** to analyze consumer feedback, track brand image, and assess ad campaign perception.
- IIM Calcutta – GenAI Impact Study:** Studied **Generative AI** effects on student **productivity, cognitive load, and guilt perception** using surveys and statistical correlation analysis.

POSITIONS OF RESPONSIBILITY

- Google Developer Student Club VIT Chennai** Social Media Lead
 - Aug 2023 - Aug 2024* Chennai, India
 - Led the social media outreach team, organizing events to grow the community’s online presence.
 - Recruited new members and assisted in their integration into the community.
- University Teaching Assistant – Database Systems Lab** Vellore Institute of Technology
 - Aug 2024 – Jan 2025* Chennai, India
 - Assisted in conducting weekly laboratory sessions for undergraduate students, focusing on SQL, relational algebra, and database normalization.
 - Evaluated assignments, supported student queries, and coordinated with faculty to ensure smooth academic delivery.